Fiscal Year:	FY 2018	**Task Last Updated:**	FY 09/08/2017
PI Name:	Radlinska, Aleksandra Ph.D.		
Project Title:	Microgravity Investigation of Cement Solidification (MICS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-1408	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Scheetz, Barry Ph.D. (Pennsylvania State University)		
Grant/Contract No.:	NNX17AC48G		
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Task Description:	The intent of this work is to utilize the microgravity environment aboard the International Space Station (ISS) to fully investigate the complex process of cement solidification. Microstructural development of hydrating cement occurs in stages during the hydration reaction and hardening process, which results in elaborate combinations of amorphous and crystalline phases. The morphology, volume fraction, and distribution of these phases ultimately determine the hardened cement’s material properties. Minimizing gravity-driven phenomena, such as thermosolutal convective flow and sedimentation, will ensure crystal growth strictly by diffusion and a considerably different microstructure than that observed in typical laboratory conditions on Earth.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed microgravity experiments will satisfy two major objectives. First, the microstructure and material properties of the benchmark samples will be evaluated; different responses to thermal and mechanical loading are expected and will be characterized in detail. Any positive attributes found in the special environment will be reported with the intent of incorporating and improving Earth-based cement and concrete processing. Given the magnitude of cement usage worldwide, even a slight change in manufacturing or usage can have huge implications on the sustainability of the cement industry.
Task Progress:	Task 1: Ampule Design and Flight Preparation. Preliminary tests were completed on ground in order to develop an experiment suitable for the International Space Station (ISS) environment, where the safety of staff and preservation of equipment is of primary importance. The experiment requires preparation of 3-compartment ampoules designed to store liquid phase, powder phase, and alcohol phase, all three to be mixed while in microgravity environment. The simplicity of the experiment is true when the experiment is conducted on Earth, but in microgravity environment extra considerations need to be taken into account, including the choice of appropriate mixing tools, ease of obtaining a homogeneous mixture, level of containment of mixing containers and their suitability for mixing a powder phase. Three different types of containers were tested at Penn State and Marshall Space Flight Center laboratories, with the objective of designing the system that is relatively simple and small in size: 1.1. MixStix Microgravity Mixing Tubes: malleable silicon tubes (=1.2 cm) that have the ends closed with plugs and compartments separated by clamps. MixStix were used in experiments with liquids in over 500 payloads launched to date on the ISS, so its safety has been proven. However, mixing a fine powder with a liquid phase to obtain a homogeneous mixture in MixStix proved to be rather difficult, due to cement powder concentrating at the nearby plug area and its limited ability to mix manually, by moving the mixing material inside the tube (due to the tube thickness). 1.2. Syringes with blunt tip needles – This set up was explored as rather safe in terms of leakage, but as in case of MixStix, mixing dry cement with a liquid for a homogeneous mixture requires substantial agitation, as the walls of syringes are rigid. In addition, the set of two 100 ml and one 10, 30, or 50 ml syringe (amounts needed for the experiment) occupies considerable volume. 1.3. Burst Pouches: as of now, the burst pouches seem to be the most appropriate setup to be used to mix cement and water on the ISS. The pouches allow manual mixing and occupy small volume. The plastic bags are continued to be tested under vacuum, to confirm that the heat seal created in the laboratory will not allow any leakage of its content (water, isopropyl, and other liquid chemical admixtures). So far, all the tests performed have been satisfactory. It should be also mentioned that 2- and 3-compartment pouches are being designed to meet all the experimental needs, which includes appropriate volumes of the compartments, adequate quality of seals, angles, and rounded edges. Task 1b: Materials characterization In preparation for the flight, all materials need to be gathered and characterized to determine their exact chemical composition. In addition to samples prepared for the flight, materials for companion samples will be set aside and proportioned into the mixing packets. This will enable preparation of the same number and type of mixtures between ground- and space-based experiments. Detailed characterization of raw materials is an essential step, as it allows critical understanding of the reactivity and materials role in microstructural development of hydrating cement paste. The tests conducted so far can be listed as follows: 2.1. X-ray fluorescence (XRF): the test provides the oxide composition of the materials to be used in the experiment. Later on, the data obtained can be related to phase diagrams, porosity, and degree of hydration of cement pastes mixed in both Earth and microgravity environments. The composition of all materials planned to be used are determined as of now. 2.2. X-ray diffraction (XRD): the test was conducted to determine the phases present in one of the powders used. 2.3. Laser diffraction: the test was conducted to provide the particle size distribution of one of the admixtures used. 2.4. Mass-loss measurements: the test was conducted for calcium-silicate suspensions in order to determine their water content. This was necessary to adjust the mixture proportions in terms of the water-to-cement ratio. Task 2: Defining Mixing Proportions. The mixing proportions (e.g., water-to-cement ratio) of each series proposed to be tested were evaluated and determined. This step was necessary to assure that a homogeneous mixture can be prepared in space and that material proportions remain within manufacturer’s recommendations. Task 3: Mechanical testing of hardened samples and other pre-flight tests. 3.1. Temperature profile of hydrating cements: The hydration of portland cement is an exothermic reaction, i.e., heat is liberated when cement powder gets in contact with water. Measuring how much heat is released during the mixing is important for the accuracy of the experiment. The task was performed by including thermocouples within the mixed material inside the pouches and measuring temperature development over time. This was performed for type I-II Portland cement. 3.2. Compressive strength (ASTM C109): the strength test of mortar cubes made with the same proportions of the proposed mixtures to be tested on ISS was conducted. Usually, the 7 and 28-day compressive strength is the most relevant mechanical property used among cement and concrete practitioners. 3.3. Amount of water required for the normal consistency of hydraulic cement paste (ASTM C187): This test is required in preparation for ASTM C191 – Time of Setting of Hydraulic Cement Paste. Knowing the time of set for each mixture is important, since the hydration of most of the mixtures launched will be arrested at specified time intervals. This will allow capturing different stages of microstructural development. Task 4: Petrographic and Microstructural analysis.

	Scanning electron microscopy (SEM): Fractured samples of the mixture containing one of the admixtures considered were prepared and analyzed under SEM. Task 5: Computer Simulations. Collaboration with NIST (National Institute of Standards & Technology) continues. Task 6: Visuals and Reports Preparation: Preparation for Science Concept Review (SCR) and Science Review Document (SRD) development. In addition to the laboratory tasks, a vast literature review was done in preparation for the SCR. Among the topics covered are: (i) modeling cement hydration through HydratiCA, developed at NIST, (ii) the most recent progresses in understanding cement hydration (i.e., direct 3D measurements of dissolving / precipitating cement particles), (iii) studies done in parallel on terrestrial and microgravity conditions, which looked at 10 seconds of cement hydration during parabolic flights, and (iv) reviewing the role of fillers, superplasticizers, and other chemicals that compose the proposed test matrix. The SRD was developed and presented to External Panel. The SCR was hosted at Penn State University on August 24th, 2017, with feedback provided by the Panelists and by the NASA Engineering Team. Recommendations from the Panelists and NASA Team are now being implemented.
Bibliography Type:	Description: (Last Updated: 10/02/2023)
Abstracts for Journals and Proceedings	Neves J, Radlinska A, Scheetz B. "Microgravity investigation of cement solidification." Presented at Coalition for Aerospace and Science (CAS) Poster Exhibition Reception, Washington, DC, June 14, 2017. Coalition for Aerospace and Science (CAS) Poster Exhibition Reception, Washington, DC, June 14, 2017. , Jun-2017